Molecular Simulation

Computer Lab & Report Instructions

Flemingsberg Campus - T41 - 30 January 2025
Groups 1-10 13:00 - 15:00
Groups 11-20 15:00 - 17:00

Alessandra Villa - KTH - Royal Institute of Technology
Maryam Majdolhosseini - KTH - Royal Institute of Technology

Computer Lab - Group Task (2 persons in each group)

Simulating the dynamic behaviour of **40 Argon atoms** in **2 dimensions** for **50 ps** at a temperature of **300 K** in a box with a length of **4 nm** in a periodic boundary condition and a time step of **0.01 ps**, using the given Python script in Canvas page.

Argon is a noble gas and has an atomic mass of **39.95 u.m.a.** The Argon atoms interact with van der Waals interactions, which we model using the Lennard-Jones **potential**:

$$V_{VDW}(r_{ij}) = 4\varepsilon\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right]$$

The parameter ε is the well-depth, and the parameter σ is the distance at which the potential is zero. **Set σ equal to 0.34 nm and ε equal to 0.979 kJ/mol.**

Address the following questions in the report using the information above and the Python script. Take care that the text is clearly written, covering all the details. Max 1500 words.

Use captions to describe the content of each figure, labels for axis definition, and legends for set definition. Report values together with units.

NOTE: Keep all the versions of the Python script in case they are requested during the corrections.

The report should be uploaded to Canvas as soon as possible and before **14th February 2025.**

Points to address in the report

a) The script uses a simple algorithm to solve Newton's equations of motion for a set of 40 interacting particles.

- Write the expression for the position x(t) and velocity v(t) for the algorithm in the script.
- An alternative algorithm is due to leap-frog. Write the expression for the position x(t) and velocity v(t) in the leap-frog algorithm.
- Discuss possible drawback of the two algorithms

b) The atoms interact with van der Waals interactions, which we model using the Lennard-Jones potential. As was explained before, the parameter ε is the well-depth. Perform a simulation using a Lennard-Jones potential with a depth of 1.5 kJ/mol (keep the initial σ value) and compare the results with the simulations you have already performed for Argon atoms. In particular:

Draw the potential energy curve in function of the distance and compare the shapes. Add a plot with both curves in it and explain what has changed.

Describe the changes in the energies derived from the Python simulations. Support your analysis with suitable graphs.

c) Total energy ($E_{tot}=E_{pot}+E_{kin}$) in the simulation should be conserved, so monitoring this during the simulation is good practice. Describe what happens to the total energy if you use a time step larger than 0.01 ps. What is the largest time step that one can use to simulate the Argon atoms and keep energy conservation. Support your results with simulation outputs.

d) Describe how temperature and mass affect the kinetic energy. To achieve this, perform simulations using different values for temperature and mass. Tips: change only one value in each simulation.

e) Finally, imagine that you want to simulate the conformation rearrangement of a small protein (a system of <100000 atoms) in detail. (the time scale of the process is nanoseconds):

- Which simulation method (Monte-Carlo or Molecular Dynamics) will you use? Why?

- Which molecular model will you use (atomistic or coarse grained model)? Why?